Element K

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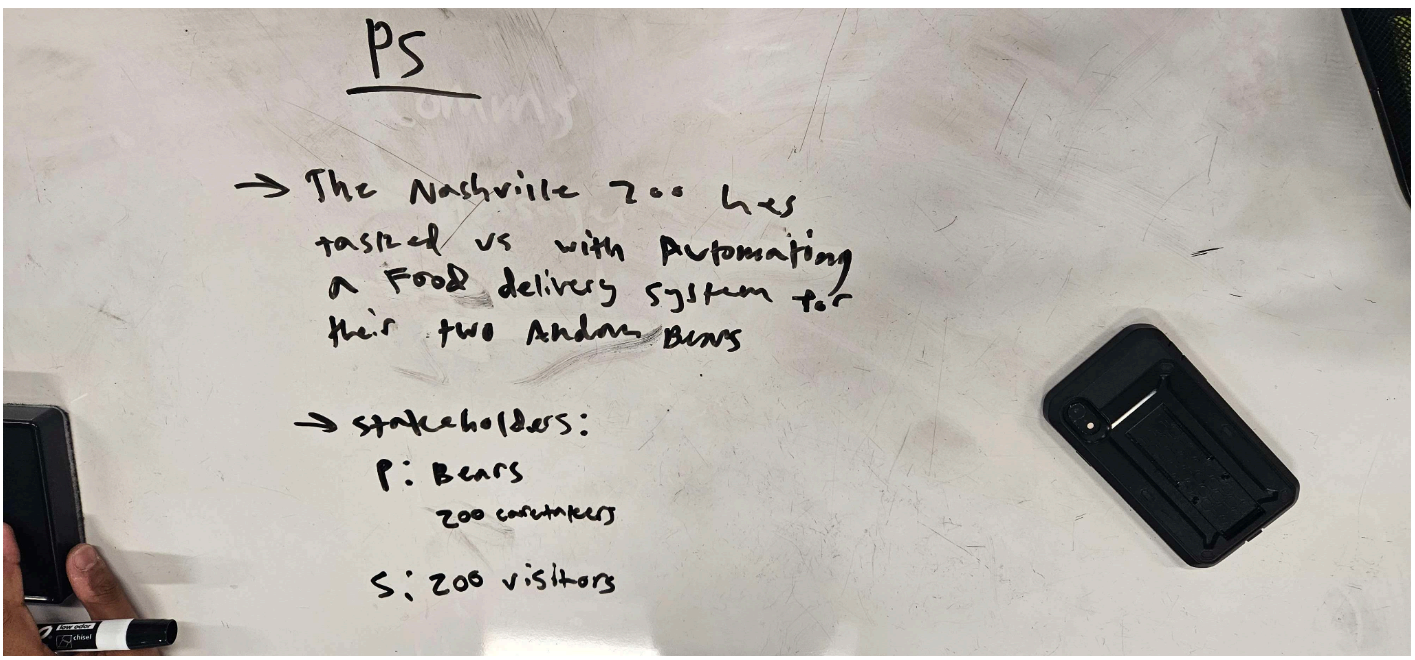
Reflection of the Design Process:

When reflecting on our overall design process, we can split up the process itself into four main parts: **Defining the Problem, Brainstorming/Prototype Selection, Prototype Building, and Expert Evaluation.**

Defining the Problem:



As shown, in the image above, the first things we did were define the problem as well as the stakeholders, both primary and secondary. We ultimately agreed that the problem that needed to be solved was the ability to move food across the Nashville Zoo's gap between the workers side to the bears side.

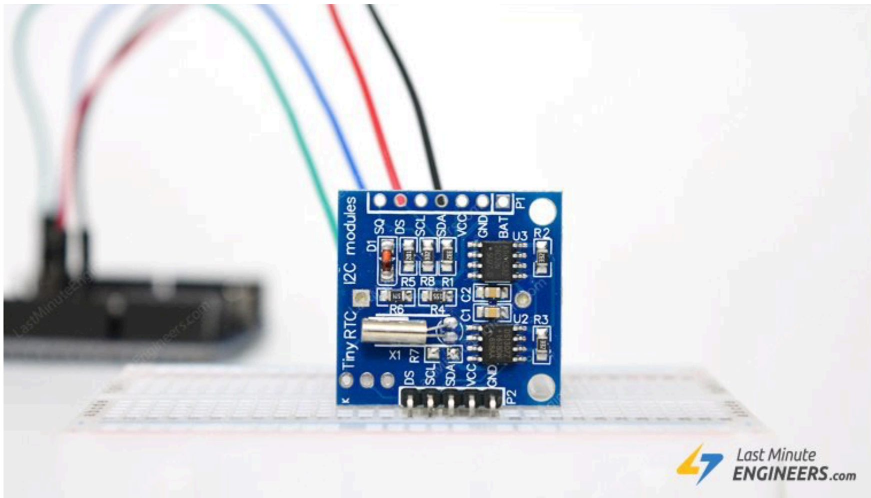
Brainstorming and Prototype Selection:

Raspberry Pi vs Arduino

Raspberry Pi	Arduino
Microcomputer	Microcontroller
Needs an operating system	Does not need an operating system
Complicated	Simple
Video out, Camera, Ethernet ports, Wifi, Bluetooth, USB, I2C, SPI, UART etc. on board	USB only for power and serial in/out, I2C, SPI, UART
Best for general computer	Best for small tasks that constantly repeat
Capable of performing a huge range of tasks	Optimised for sensing and controlling the world around it
Best for more advanced makers	Best for beginners
Programmed in many languages, including C/C++, Python, Ruby	Programmed in C/C++
Relatively high power consumption	Relatively low power consumption

Raspberry Pi Full Stack Tech Explorations

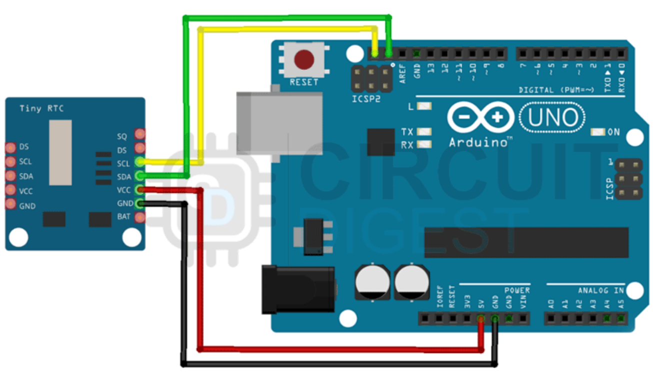
When it came to choosing the brains for our project, little brainstorming was needed and we knew that it came down to whether we wanted to use an arduino or a raspberry pi. Both of these devices are very small and lightweight, as well as they contain the intelligence needed for our project to succeed.Through research in the forms similar to the image shown above, we concluded that arduinos are much more optimal for in the field work, and that the raspberry pi is more of a microcomputer used for software and back end development. Additionally, we knew we would need some way of keeping track of the time, which is why we obtained an Arduino RTC, or real time clock module. This module is very useful as it allows for the Arduino to keep track of time and utilize it in code.



With this piece, we were now able to create essentially a small clock that could keep track of time and thus allow the arduino to perform actions at a randomly generated time.

Prototype Building:

Regarding our prototype building, the mechanical side of it was easy enough to assemble as shown in the image below.



This image shows how the arduino and rtc module would be setup in order for the rtc to track the time. Then, we were able to create some simple software, with the help of the library RTClib, to randomize the time at which the code shoots.

```
void loop()
{




  DateTime now = rtc1.now();
  now.dayOfTheWeek();
  if((now.hour() == goalTime[0]) && (now.minute() == goalTime[1]) && (now.second() ==
30)){
    //run the special code for launch
    lcd.display("Team Geppetto's Automation!");
  }
}
```

The code of the loop (as pasted off of our github) shows the simple setup where at a random time, it prints the line to the LCD display, although that code was to be replaced with the code that was spun the motors (refer to element J for code)

Expert Evaluation/Takeaways

Overall, after speaking with our two main experts, that being Ms. Daniel on the zoo's side as well as Mr. Rodriguez on the software side, we came away with a few things

- The amount of times the launcher is used is actually very minimal, and the normal meals that the bear eat are hand fed
    - This means that the bear snacks are what are sent out of the launcher
- The bears also have a certain days where they are hand fed snacks as a exhibit for the zoo, and thus that lowers the amount of times that the launcher is under load even more than previously
- Regarding our code side, using a hashmap was definitely a good decision as in terms of organizing data, it is much better than something like a 2D array, which is the alternative method we had for storing the feeding times
- The randomization part of the algorithm turned out to be much harder than expected, as finding out what day of the week it was meant we had to generate our random times in our continuous loop, rather than in the setup of the code

Overall, we had many takeaways from the full engineering design process than we expected, and the biggest thing was how we had to scale up the project, as new feedback from people came in and gave us newer and more specific criteria to fufill.